Robust but Overconfident: Corruption-Robustness Fine-Tuning Erodes Visual Abstention in Vision–Language Models

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Abstract

*Vision–language models (VLMs) are deployed precisely where images are hardest to read: assistive photography for blind users, perception through rain, fog and motion blur, and low-quality scans. In these settings the safe response to an uninterpretable input is not a confident guess but an admission that the image cannot support an answer. We show that the obvious way to prepare a VLM for degraded inputs—fine-tuning it for corruption robustness—*silently *removes that behavior. Using two standard adaptation recipes, on-policy reverse-KL distillation (RKL) and Group Relative Policy Optimization (GRPO), we find that corruption-robust training improves clean and corrupted accuracy and transfers* zero-shot *to corruption types never seen in training, yet drives both models to answer uninterpretable and image-free inputs far more often than the base model. The regression is invisible to accuracy: the most accurate model we train is the worst at recognizing when it cannot see. We then explain the failure. A linear probe recovers answerability from frozen hidden states at* $\mathrm{AUROC} \approx 0.93$ *in every model, so the ability is suppressed, not erased; and a reasoning-transplant test shows the two recipes break for* opposite *reasons—RKL cannot reason its way to "unanswerable," whereas GRPO reasons correctly but has learned an answer-level shortcut that commits regardless. Finally we recover abstention two ways—adding* 10% *unanswerable examples to training, and inference-time activation steering toward the base model—each at no cost to the corruption gains, with the training objective deciding which works better. The results argue that corruption robustness and calibrated abstention must be measured, and optimized, together.*

1. Introduction

Modern vision–language models are most useful in exactly the conditions that most degrade an image. Assistive applications answer questions about photographs taken by blind and low-vision users, who cannot check whether a shot is focused, framed, or lit; perception systems in vehicles must see through rain, fog, glare and motion; and scientific or clinical tools routinely operate on noisy scans. What unites these settings is that a confident wrong answer is the most damaging possible output, because whoever acts on it has no signal that it is wrong. The correct behavior when an input is uninterpretable is *abstention*: saying that the image does not support an answer.

The natural way to make a VLM robust to such inputs is to fine-tune it on corrupted images, and two paradigms dominate. *Knowledge distillation* trains a student that sees a degraded image to imitate a larger teacher that sees the clean one; it is inexpensive, needs no labels, and uses only teacher forward passes. *Reinforcement learning with verifiable rewards*, here Group Relative Policy Optimization (GRPO) [11], rewards the model when its answer matches the gold answer. Both reliably raise accuracy on corrupted inputs. Our central finding is that both also do something their accuracy never reveals: **they teach the model to stop abstaining.**

We establish this on Qwen3-VL-4B-Instruct [1], fine-tuned with on-policy reverse-KL distillation (RKL) [3] and with GRPO on 30k GQA questions [8] corrupted by a 16-style engine, and evaluated on corrupted TextVQA [12], GQA [8] and VQAv2 [2], together with VizWiz [4]: real blind-user photographs, a third of whose questions are genuinely unanswerable. We make four contributions.

- **A silent abstention regression.** Corruption-robust fine-tuning raises clean and corrupted accuracy—with gains that transfer zero-shot to unseen corruptions (§4.1, Appendix A)—yet answers unanswerable VizWiz questions and image-free inputs far more often than the base model (§4.2). GRPO, the most accurate model, is the worst abstainer, and no accuracy metric exposes this.
- **The ability is suppressed, not erased (RQ1).** A linear probe decodes answerability from frozen hidden states at $\mathrm{AUROC} \approx 0.93$ in all three models and still flags most of the inputs they wrongly answer; the failure is behavioral,

not representational (§4.3).

- **Two opposite failure mechanisms (RQ2).** A reasoning-transplant test shows RKL fails on the *reasoning* side (given a "can't-tell" chain it abstains 98.4% of the time, so unaided it never reaches that chain), while GRPO fails on the *answer* side (given the same chain it still answers 91.8% of the time, a reward-induced shortcut) (§4.4).
- **Two mechanism-matched remedies (RQ3).** A 10% unanswerable data mix and inference-time activation steering each restore abstention without surrendering the corruption gains, and the objective decides which is better—entropy-gated steering for RKL, the missing data signal for GRPO (§4.5).

The lesson is methodological: corruption robustness and calibrated abstention are coupled, standard accuracy benchmarks hide the coupling, and the right fix depends on how the model was trained. We therefore report an explicit abstention axis beside accuracy throughout.

2. Related Work

VLMs and visual question answering. Instruction-tuned VLMs on strong language backbones are now the default interface for image understanding [1]. They are measured on VQA benchmarks targeting compositional reasoning (GQA [8]), language-prior control (VQAv2 [2]), and scene-text reading (TextVQA [12]). VizWiz [4] is unique for our purpose: its images are taken by blind users, are often unreadable, and a large share of its questions cannot be answered, making it the natural testbed for abstention under real degradation.

Corruption robustness. ImageNet-C [5] established common corruptions—noise, blur, weather, digital artifacts at graded severity—as the standard average-case robustness benchmark, distinct from worst-case adversarial robustness. We adopt its corruption families for evaluation but train with a *different* corruption engine, so most test corruptions are unseen and the measured gains reflect generalization rather than memorized pixel statistics.

Distillation and RL for adaptation. Knowledge distillation [6] transfers a teacher's behavior to a student; for generative models, MiniLLM [3] shows that *reverse* KL is preferable to forward KL because it is mode-seeking and avoids placing mass on regions the teacher never visits. Our distillation is on-policy reverse-KL with a clean-image teacher and a corrupted-image student. On the RL side, GRPO [11] replaces PPO's value network [10] with group-relative advantages from sampled rollouts, and is a common recipe for verifiable-reward fine-tuning; LoRA [7] is a standard efficient variant of both. A recognized hazard of reward-based training is *shortcut* behavior; we give a causal demonstration of one such shortcut—always answering—in the abstention setting.

Abstention and activation steering. Knowing when not to answer is central to reliability, and VLMs are prone to confident answers on unanswerable or image-free prompts. Activation- and representation-engineering methods edit internal activations at inference time to modulate a target behavior; Contrastive Activation Addition (CAA) [9] forms a steering vector from the mean difference of residual-stream activations over positive/negative pairs and adds it during generation, on top of fine-tuning rather than undoing it [13]. We build contrastive vectors from base-vs-trained response pairs to push a fine-tuned model back toward the base model's abstention, and add an entropy-gated variant matched to distillation's reasoning-side failure.

3. Method and Experimental Setup

3.1. Corruption-robust fine-tuning

Backbone and data. The student is Qwen3-VL-4B-Instruct [1]; the distillation teacher is the frozen Qwen3-VL-8B. Training uses 30,000 questions sampled (seed 0) from GQA `train_balanced` [8] over 22,607 distinct images. Each question receives one of 16 corruption styles—lighting, camera motion, occlusion, weather, sensor noise and digital compression—balanced at 1,875 questions per style and applied at severity $4/5$ (Appendix B, Fig. 2). Both methods train for one epoch (3,750 steps, batch size 8).

Reverse-KL distillation (RKL). Let $\tilde{x}$ be a corrupted image, x its clean counterpart, c the prompt and y a response. The student q_θ sees $\tilde{x}$ and is distilled toward the frozen teacher p_ϕ, which sees the clean image x. We minimize the token-level *reverse* KL on-policy, sampling $G{=}2$ rollouts $y \sim q_\theta(\cdot \mid \tilde{x}, c)$ per prompt:

$$\mathcal{L}_{RKL}(\theta) = \mathbb{E}_{\tilde{x},x,c,\, y\sim q_\theta}\left[\sum_t \mathrm{KL}\big(q_\theta^{(t)} \,\|\, p_\phi^{(t)}\big)\right], \quad (1)$$

with $q_\theta^{(t)} = q_\theta(\cdot \mid y_{<t}, \tilde{x}, c)$ and $p_\phi^{(t)} = p_\phi(\cdot \mid y_{<t}, x, c)$. Reverse KL is mode-seeking: the student concentrates on the teacher's dominant modes [3]. Distillation needs no gold answers and only teacher forward passes.

Group Relative Policy Optimization (GRPO). GRPO sees only the corrupted image. For each question it samples $\{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid \tilde{x}, c)$ with $G{=}8$ and assigns an exact-match reward $r_i = \mathbb{1}[\mathsf{answer}(o_i) = a^\star]$ against the GQA gold answer $a^\star$. Advantages are group-normalized

and the policy is updated by the clipped surrogate with a KL anchor:

$$\hat{A}_i = \frac{r_i - \mathrm{mean}(\mathbf{r})}{\mathrm{std}(\mathbf{r}) + \varepsilon}, \tag{2}$$

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}\Big[\tfrac{1}{G}\sum_{i=1}^{G}\tfrac{1}{|o_i|}\sum_t \min\big(\rho_{i,t}\hat{A}_i,\ \bar{\rho}_{i,t}\hat{A}_i\big) - \beta\,\mathrm{KL}(\pi_\theta\|\pi_{\mathrm{ref}})\Big], \tag{3}$$

where $\rho_{i,t} = \pi_\theta(o_{i,t} \mid \tilde{x}, c, o_{i,<t})/\pi_{\theta_{\mathrm{old}}}(o_{i,t} \mid \tilde{x}, c, o_{i,<t})$ and $\bar{\rho}_{i,t} = \mathrm{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon)$. Because the reward exists only for answerable GQA questions, nothing in Eq. (3) ever rewards abstaining—a signal gap we return to in §4.5.

3.2. Evaluation and the abstention axis

TextVQA, GQA and VQAv2 are corrupted with the ImageNet-C [5] engine (19 types, 4 groups, severities 1–5): one corrupted copy per question, drawn to balance the 95 type×severity cells, with every question also asked on its clean image. Only 8 of the 19 types have a training counterpart; the remaining 11 (∼58% of corrupted questions) are *unseen*, so robustness on them is zero-shot (Appendix A). VizWiz [4] adds no synthetic corruption—its 2,934 answerable and 1,385 unanswerable questions carry real degradation—and is our primary abstention benchmark. We track three abstention measurements:

- **VizWiz-unanswerable accuracy** (higher is better): correct abstention on genuinely unanswerable questions.
- **No-image accuracy** (lower is better): accuracy when the image is withheld; answering means guessing from language priors.
- **Refusal rates**: over-refusal on answerable questions (lower is better) and correct abstention with no image (higher is better).

3.3. Diagnostics

Answerability probe (RQ1). For each VizWiz item we take the last-prompt-token hidden state $h^{(\ell)}$ at every layer ℓ, before generation. A logistic-regression probe $\hat{y} = \sigma(w_\ell^\top h^{(\ell)} + b_\ell)$ is trained on the VizWiz training split to predict answerable vs. unanswerable and tested on validation; a shuffled-label probe is the control.

Reasoning transplant (RQ2). We take unanswerable questions the trained model answered but the base model refused (its chain ends in "can't tell"), prepend that base chain to the trained model, and measure whether it now abstains—isolating whether the failure is in *producing* the reasoning or in *acting* on it.

3.4. Interventions (RQ3)

Unanswerable data mix. We corrupt 3,334 further GQA questions past the usable range (severity 100, or 50 for snow) until nothing is legible, relabel them "unanswerable," and append the note *"If the image does not contain enough information to answer, the answer is unanswerable."* With the 30k this forms a 10% mix (33,334 rows), trained for the same 3,750 steps; for RKL the teacher also sees the destroyed image (Appendix B).

Activation steering. At inference we add a steering vector to the residual stream at layer 18 of 36, on generated tokens only:

$$h_t^{(18)} \leftarrow h_t^{(18)} + \lambda\, v_{18}. \tag{4}$$

Both vectors come from calibration data on training splits (no evaluation overlap). The *image-contrast* vector uses VizWiz-train unanswerable photos where the base model abstained and the trained model did not, taking the base−trained hidden-state difference over the trained model's response tokens:

$$v_\ell^{\mathrm{con}} = \underset{\text{resp. tok.}}{\mathrm{mean}}\Big[h_\ell^{\mathrm{base}}(x) - h_\ell^{\mathrm{trained}}(x)\Big], \tag{5}$$

where x is the image, the question and the trained model's own response. The *no-image* vector uses image-free calibration questions where the base refuses and the trained model answers (150 pairs/dataset):

$$v_\ell^{\mathrm{noimg}} = \underset{\text{pairs}}{\mathrm{mean}}\Big[\bar{h}_\ell(q{+}\text{refusal}) - \bar{h}_\ell(q{+}\text{answer})\Big], \tag{6}$$

where $\bar{h}_\ell$ averages layer-ℓ states over response tokens; the refusal is the base model's, the answer the trained model's. We set λ=8 (contrast) and λ=1 (no-image), each rescaled to a reference norm. Steering is *always on* or *entropy-gated*: applied only where the next-token predictive entropy signals confusion,

$$\text{steer at } t \iff H_t = -\sum_v p_\theta(v)\log p_\theta(v) > \tau,\quad \tau = 0.5. \tag{7}$$

Contrast pairs and vector counts are in Appendix C.

4. Results

We first establish the headline trade-off, then answer three research questions.

4.1. Robustness, and that it transfers zero-shot

Table 1 reports clean and corrupted accuracy. GRPO is strongest on every dataset, raising corrupted GQA by +10.2 and corrupted VQAv2 by +9.5 over the base model while also improving clean accuracy; RKL improves the corrupted domain (e.g. +4.2 corrupted VQAv2) and holds

clean accuracy flat. These gains are not an artifact of the training engine. Averaged over the 19 evaluation corruptions (Appendix A, Fig. 1), GRPO improves *unseen* corruptions by $+8.7$ and seen ones by $+8.2$, and RKL by $+1.9$ and $+2.2$ respectively: the robustness transfers *zero-shot*, GRPO's unseen gain even exceeding its seen gain.

4.2. The silent abstention regression

The same training degrades abstention. VizWiz-unanswerable accuracy (Table 1) falls from 76.0 (base) to 60.3 (GRPO) and 56.5 (RKL). The no-image test is sharper (Table 2): with the image withheld, base accuracy sits near the floor (1.9–14.5), but the fine-tuned models answer anyway, GRPO reaching 46.0 on VQAv2 and 36.3 on GQA. **The most accurate model is the worst abstainer**—and nothing in the accuracy columns of Table 1 would reveal it.

4.3. RQ1: Erased, or only suppressed?

A linear probe on frozen hidden states separates answerable from unanswerable VizWiz questions at $\mathrm{AUROC} \approx 0.93$ in all three models, far above the shuffled-label control (Table 3): fine-tuning does not erase the *representation* of answerability. Among the unanswerable questions the models wrongly answered, the probe still flags 76–77% as unanswerable (711 items for RKL, 765 for GRPO). The models internally "know" these inputs are unanswerable, yet answer them. The lone exception is image-missingness: with the image withheld (Table 4) the flag rate collapses for RKL (GQA $68\% \to 31\%$), so distillation has partly lost its sense that the image is gone, while GRPO keeps it. The failure is thus one of *behavior*, not *knowledge*.

4.4. RQ2: Do the two recipes fail the same way?

They fail oppositely, and the reasoning-transplant test (Table 5) shows it. Handed a base "can't-tell" chain for a question it had answered, RKL abstains 98.4% of the time: give it the right reasoning and it behaves correctly, so its own failure is that it never *produces* that reasoning. Consistently, RKL's responses on these items are far longer (median 341.5 tokens vs. 109 for base and GRPO on the same selection; Appendix D, Fig. 8)—the signature of an unsure, high-entropy chain that talks itself into an answer. GRPO, given the identical chain, still answers 91.8% of the time: its reasoning is sound, but it has learned an *answer-level shortcut*, committing to a definitive answer no matter what the chain concludes, because exact-match reward always paid off for answering. Figure 8 makes this vivid: on an illegible oven dial, GRPO's own reasoning states the setting "cannot be answered," then emits "350." In short, **RKL cannot reason its way to unanswerability; GRPO reasons there but will not act on it.**

4.5. RQ3: Recovering abstention without losing the gains

A 10% unanswerable mix. Adding 10% unanswerable data restores abstention for both objectives while clean and corrupted accuracy hold or improve (Table 6). VizWiz-unanswerable rises $56.5 \to 66.9$ for RKL and $63.5 \to 69.7$ for GRPO, and no-image accuracy drops for GRPO across the board (Table 7: GQA $36.3 \to 30.7$, VQAv2 $46.0 \to 36.5$). The abstention failure is therefore not an unavoidable cost of robustness—it is a *missing training signal*, and a small amount of it suffices.

The two objectives respond differently, confirming the RQ2 mechanism. As refusal rates (base over-refusal 12.8%, base no-image refusal 98.3%), the mix moves GRPO's no-image refusal $16.0\% \to 84.6\%$ and its unanswerable refusal $51.9\% \to 92.8\%$—the dramatic recovery a pure signal-gap predicts—at the cost of some over-refusal on answerable questions (29.4%), the same RL optimization pushing too far once a reward direction exists. RKL recovers less raw abstention and still under-abstains with no image, but what it recovers is *calibrated* (over-refusal 14.1%, unanswerable refusal 72.6%, both near base), consistent with an inherited, over-committing reasoning style that a small data fraction only partly rewrites. One intervention lands the two models in different places—proof that their failures are mechanistically distinct.

Activation steering. Steering recovers more abstention than the data mix, but must be matched to the objective (Tables 8–9). For RKL, which fails on the reasoning side, *entropy-gated* steering—active only where the next-token entropy signals confusion (Eq. 7)—lifts VizWiz-unanswerable $56.1 \to 64.1$ with clean and corrupted accuracy essentially unchanged; the stronger always-on image-contrast vector reaches 90.3 but at a real accuracy cost (clean GQA $55.4 \to 46.4$). For GRPO, which fails on the answer side, the no-image vector raises VizWiz-unanswerable $60.1 \to 81.7$ and nearly restores base-level no-image refusal (no-image accuracy GQA $36.5 \to 21.5$, VQAv2 $45.8 \to 16.1$) with negligible accuracy loss. GRPO must be steered harder than RKL to reach the base model's abstention, and beyond a point steering trades away the corruption gains—mirroring, at inference time, the data-mix overshoot.

5. Discussion and Conclusion

A common way of training VLMs for corruption robustness quietly makes them worse at admitting when they cannot see; standard accuracy benchmarks miss the regression; and it is recoverable. The failure is behavioral, not representational (RQ1): answerability stays linearly decodable after fine-tuning. The two dominant recipes fail for opposite

Table 1. **Clean and corrupted accuracy, and VizWiz abstention.** Corruption-robust fine-tuning improves accuracy (GRPO most) but both models answer far more unanswerable VizWiz questions than the base model (VizWiz-Unans., higher is better). Best per column in **bold**.

	TextVQA		GQA		VQAv2		VizWiz		
Model	Clean	Corr.	Clean	Corr.	Clean	Corr.	Unans.	Answ.	Overall
4B base	81.3	67.9	55.8	51.9	79.2	73.0	**76.0**	51.0	59.0
Distillation (RKL)	80.1	68.3	55.5	53.7	80.8	77.2	56.5	57.2	57.0
GRPO	**84.6**	**71.0**	**65.4**	**62.1**	**87.0**	**82.5**	60.3	**63.5**	**62.4**

Table 2. **No-image accuracy** (*lower is better*). With no image, the fine-tuned models answer from language priors far more than the base model.

Model	GQA	VQAv2	TextVQA
Base	**14.5**	**13.9**	**1.9**
Distillation (RKL)	31.8	41.2	9.3
GRPO	36.3	46.0	9.8

Table 3. **Answerability probe** on frozen hidden states. Answerability is linearly decodable at $\mathrm{AUROC} \approx 0.93$ after fine-tuning.

Model	Best layer	Probe AUROC	Shuffled
Base	23	0.931	0.48
Distillation (RKL)	24	0.927	0.52
GRPO	23	0.930	0.50

Table 4. **With the image withheld, share of questions the probe flags unanswerable.** RKL loses much of its sense that the image is missing.

Model	GQA	VQAv2	TextVQA	VizWiz
Base	68	75	98	100
Distillation (RKL)	31	38	94	99
GRPO	56	64	98	100

Table 5. **Reasoning transplant.** Fraction that abstains when given a base "can't-tell" chain for a question it had answered. RKL fails on the reasoning side; GRPO on the answer side.

Model	Items	Refuses w/ "can't-tell" CoT
Distillation (RKL)	308	98.4%
GRPO	389	8.2%

reasons (RQ2): distillation on the reasoning side, GRPO on the answer side via a reward-induced shortcut. And the remedy should match the mechanism (RQ3): entropy-gated steering for distillation, the missing unanswerable signal for GRPO—each at no cost to corruption gains that already transfer zero-shot. The practical message is simple but consequential: wherever a confident wrong answer is dangerous, corruption robustness and calibrated abstention must be measured, and optimized, together.

Limitations. We study one backbone family and four benchmarks; whether the two failure signatures generalize to other architectures, reward designs and corruption regimes is open. Steering uses a single layer and a fixed gate threshold; adaptive, multi-layer and per-token policies are natural extensions.

Table 6. **Effect of a** 10% **unanswerable data mix.** Abstention (VizWiz-Unans.) recovers for both objectives while clean and corrupted accuracy hold.

	TextVQA		GQA		VQAv2		VizWiz		
Model	Clean	Corr.	Clean	Corr.	Clean	Corr.	Unans.	Answ.	Overall
Base	81.3	67.9	55.8	51.9	79.2	73.0	76.0	51.5	59.0
Distillation (RKL)	80.1	68.3	55.5	53.7	80.8	77.2	56.5	57.2	57.0
+10% unanswerable	79.8	67.0	55.8	53.4	79.6	76.7	66.9	58.6	61.3
GRPO	84.6	71.0	65.4	62.1	87.0	82.5	63.5	60.3	62.4
+10% unanswerable	80.0	70.7	65.4	62.6	86.0	81.5	**69.7**	**66.0**	**67.2**

Table 7. **No-image accuracy with the** 10% **mix** (*lower is better*). GRPO's image-free guessing drops sharply.

Model	TextVQA	VQAv2	GQA
Base	**1.9**	**13.9**	**14.5**
Distillation (RKL)	9.3	31.8	31.8
+10% unanswerable	8.5	31.5	31.5
GRPO	9.8	46.0	36.3
+10% unanswerable	4.8	36.5	30.7

Table 8. **Activation steering.** Entropy-gated steering recovers RKL's abstention at no accuracy cost; a no-image vector recovers GRPO's. Aggressive always-on contrast (RKL + image contrast) buys more abstention but spends the corruption gains.

	TextVQA		GQA		VQAv2		VizWiz		
Model	Clean	Corr.	Clean	Corr.	Clean	Corr.	Unans.	Answ.	Overall
4B base	81.8	67.9	55.8	51.9	79.2	73.0	76.0	51.0	59.0
RKL unsteered	80.5	68.3	55.4	53.8	81.1	77.2	56.1	57.3	56.9
RKL + gated steering	80.9	68.5	55.7	53.7	81.3	77.5	64.1	59.1	60.7
RKL + image contrast	74.8	61.3	46.4	43.4	72.6	68.0	**90.3**	**64.0**	**72.4**
GRPO	**84.5**	71.2	**65.1**	62.0	87.2	82.6	60.1	63.8	62.6
GRPO + no-image steering	**84.8**	70.7	64.6	61.7	87.0	82.7	81.7	68.7	**72.9**

Table 9. **No-image accuracy under steering** (*lower is better*). GRPO + no-image steering returns almost to base-level abstention.

Model	TextVQA	GQA	VQAv2
4B base	**1.9**	**14.5**	**13.9**
RKL unsteered	9.2	31.9	41.3
RKL + gated steering	8.9	32.0	41.7
RKL + image contrast	3.7	21.2	30.8
GRPO	9.7	36.5	45.8
GRPO + no-image steering	1.9	21.5	16.1

Appendix
Supplementary Material

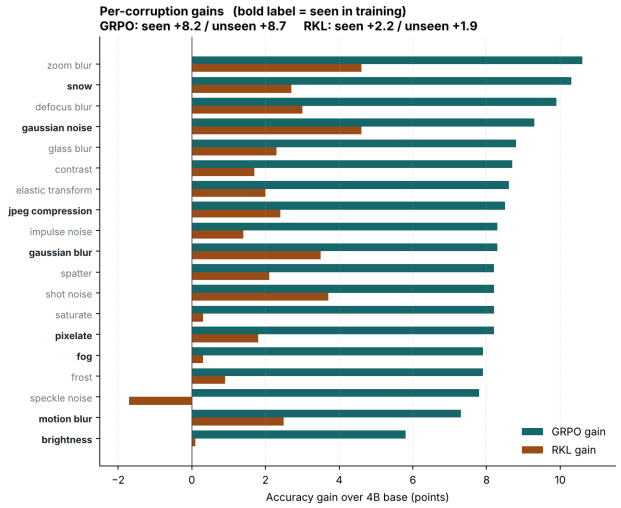


Figure 1. **Per-corruption gains** over the 4B base, sorted by GRPO gain. Bold y-labels are corruptions with a training counterpart (seen); the rest are unseen. GRPO (teal) improves all 19 types by 5.8–10.6 points regardless of seen/unseen status, evidencing zero-shot transfer.

This appendix provides the per-corruption results and zero-shot transfer analysis (§A), the full data and corruption catalogue with figures (§B), the steering-vector construction and contrast pairs (§C), and qualitative examples of the failure modes (§D).

A. Per-Corruption Gains and Zero-Shot Transfer

Table 10 lists base accuracy and the RKL/GRPO gains for each ImageNet-C evaluation corruption, and Fig. 1 shows them sorted. Because training and evaluation use different engines, even "seen" names differ in pixels, and 11 of 19 types have no training counterpart. The group means (Table 11) make the transfer explicit: GRPO's mean gain on *unseen* corruptions (+8.7) matches—and slightly exceeds—its gain on *seen* ones (+8.2), and RKL's unseen gain (+1.9) tracks its seen gain (+2.2). GRPO improves every type; the only regression is RKL on speckle noise (−1.7), an unseen noise variant.

B. Data and Corruptions

Table 12 summarizes the training, evaluation and calibration data; Figs. 2–6 show the corruption families, their severity progression, the destroyed-GQA images, and representative real VizWiz photographs. GQA `train` and `testdev` share no images, and the steering pool uses only training splits, so nothing used for training or calibration appears in evaluation. The 10% mix adds "Destroyed GQA": 3,334 questions corrupted past the usable range and relabeled "unanswerable." The seven destruction styles (Table 13, Fig. 5) were chosen because both base models almost always refused them; styles that stayed legible even at extreme strength (e.g. rotation, shadow, JPEG) were rejected.

Table 10. **Per-corruption base accuracy and gains.** "Seen" marks a training counterpart; gains are absolute points over the 4B base.

Corruption	Group	Seen	Base	RKL	GRPO
zoom blur	blur	no	49.7	+4.6	+10.6
gaussian blur	blur	yes	58.9	+3.5	+8.3
defocus blur	blur	no	55.2	+3.0	+9.9
motion blur	blur	yes	62.1	+2.5	+7.3
glass blur	blur	no	54.9	+2.3	+8.8
jpeg compression	digital	yes	63.6	+2.4	+8.5
elastic transform	digital	no	62.0	+2.0	+8.6
pixelate	digital	yes	62.2	+1.8	+8.2
contrast	digital	no	63.4	+1.7	+8.7
saturate	digital	no	61.7	+0.3	+8.2
gaussian noise	noise	yes	58.4	+4.6	+9.3
shot noise	noise	no	58.2	+3.7	+8.2
impulse noise	noise	no	58.9	+1.4	+8.3
speckle noise	noise	no	62.8	−1.7	+7.8
snow	weather	yes	58.4	+2.7	+10.3
spatter	weather	no	62.3	+2.1	+8.2
frost	weather	no	59.4	+0.9	+7.9
fog	weather	yes	63.5	+0.3	+7.9
brightness	weather	yes	66.3	+0.1	+5.8

Table 11. **Zero-shot transfer.** Mean gain over the base model on seen vs. unseen corruptions.

Split	#types	RKL mean	GRPO mean
Seen (in training)	8	+2.2	+8.2
Unseen (zero-shot)	11	+1.9	+8.7
All	19	+2.0	+8.5



Figure 2. **Training corruptions.** The clean reference image and the 16 corruption styles of the training engine at severity $4/5$. Each training question receives exactly one style; the engine spans lighting, camera motion, occlusion, weather, sensor noise and digital compression.

Table 12. **Data at a glance.** Train and eval use different corruption engines by design.

Set	Source	#Q	Corruption / role
GQA train	GQA `train_bal`.	30,000	16 styles, sev. 4 — train
Destroyed GQA	GQA `train_bal`.	3,334	7 styles, unreadable — 10% mix
TextVQA-C	TextVQA val	5,000	ImageNet-C 19×5 — eval
GQA-C	GQA `testdev_bal`.	12,578	ImageNet-C 19×5 — eval
VQAv2-C	VQAv2 val (COCO)	5,000	ImageNet-C 19×5 — eval
VizWiz	VizWiz-VQA val	4,319	real (2,934 ans / 1,385 unans) — eval
Steer calib.	train splits ×4	—	matches eval set — vectors

Table 13. **Destroyed-GQA styles** for the 10% mix, with base refusal rates at destruction strength.

Style	Sev.	#Q	4B refuses	8B refuses
gaussian_noise	100	477	100%	100%
gaussian_blur	100	477	96%	100%
motion_blur	100	476	92%	100%
rain	100	476	96%	100%
overexposure	100	476	88%	100%
night	100	476	92%	92%
snow	50	476	92%	92%

C. Steering-Vector Construction

Both vectors are rebuilt from calibration responses on training splits with no evaluation overlap. The image-contrast vector (Eq. 5) keeps a question when the base model abstained and the trained model did not; the no-image vector (Eq. 6) samples 150 pairs per dataset where the base refuses and the trained model answers. Figure 7 shows example contrast pairs: the base response (positive, abstains) and the trained response (negative, answers) that are differenced to form the vector. At evaluation, λv_{18} is added at layer $18/36$ on generated tokens only (Eq. 4), with λ=8 (contrast) or λ=1 (no-image), rescaled to a reference norm; the gate (Eq. 7) applies steering only when next-token entropy exceeds 0.5. Table 14 gives the pair counts; GRPO already refuses most no-image VizWiz questions, so only 184 qualified there.

D. Qualitative Analysis

On the 1,385 unanswerable VizWiz questions, base refusal is 76.0% against 56.5% (RKL) and 60.3% (GRPO). The two failure signatures are visible side by side in Fig. 8. On the same selection, RKL's median response is 341.5 tokens against 109 for base and GRPO—distillation produces a long, unsure chain that drifts into an answer, and often runs to the 768-token limit. GRPO instead emits a short, confident answer whose reasoning frequently *states* that the

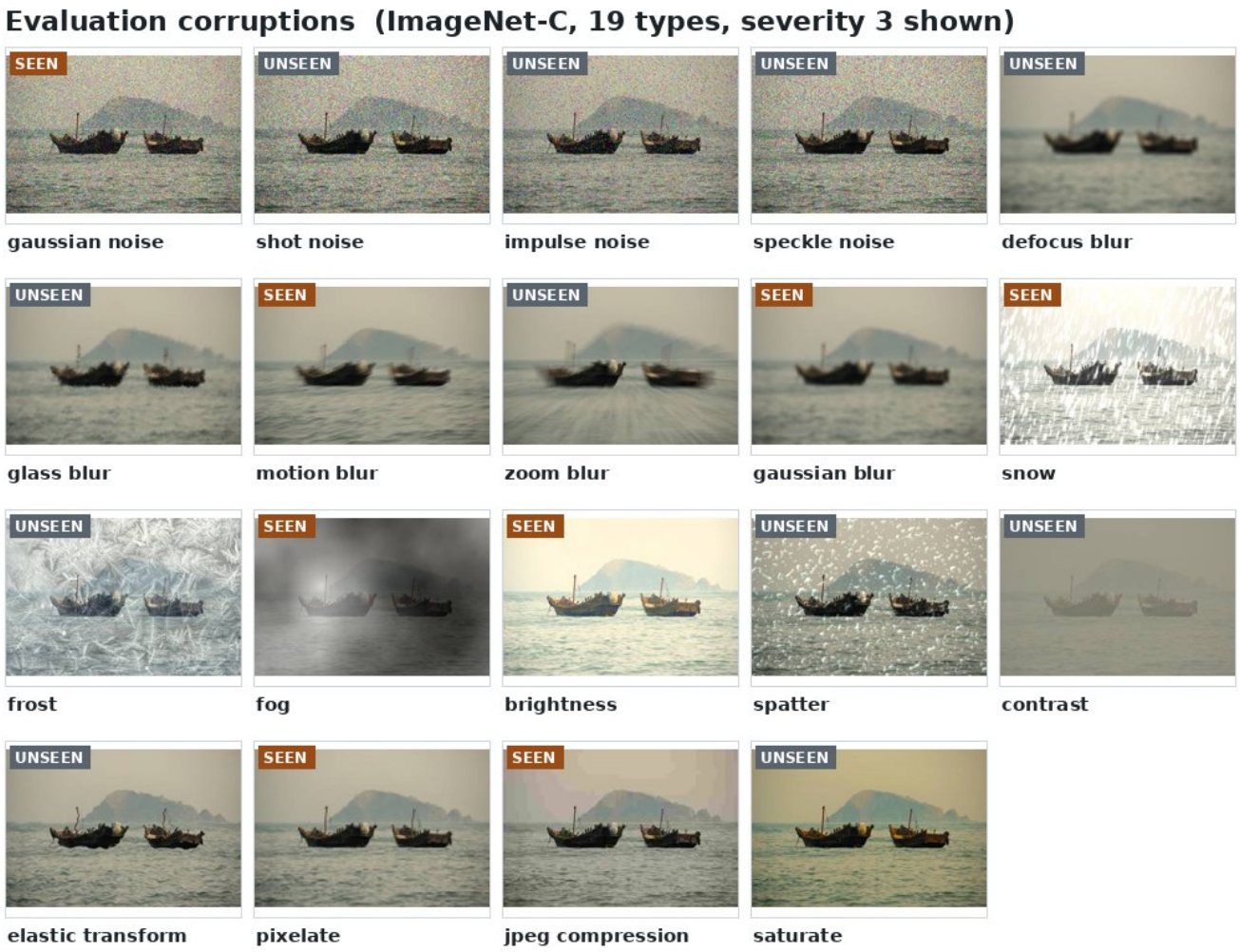


Figure 3. **Evaluation corruptions.** The 19 ImageNet-C evaluation types (severity 3 shown), tagged by whether a counterpart appeared in training. 11 of 19 are *unseen*; because training and evaluation use different engines, even "seen" names differ in pixels.

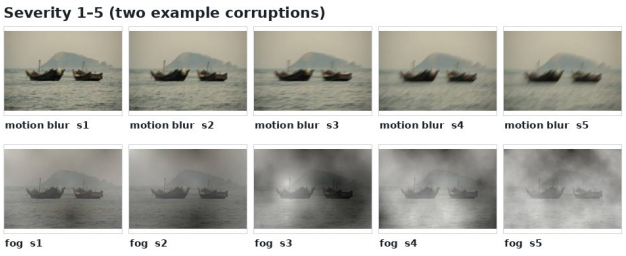


Figure 4. **Severity.** Two evaluation corruptions at severities 1 to 5; each evaluation question receives one severity, balanced across the 95 type×severity cells.

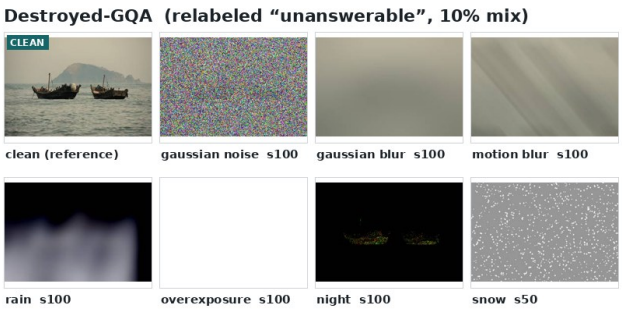


Figure 5. **Destroyed-GQA.** Seven styles pushed far past the usable range (severity 100, or 50 for snow) until nothing is legible; these images are relabeled "unanswerable" to form the 10% mix.

image is illegible: on the oven-dial example its chain concludes the setting "cannot be answered from the image,"



Figure 6. **Example inputs.** A real VizWiz unanswerable photograph (the degradation is intrinsic, not synthetic) and synthetic clean/corrupted evaluation pairs for GQA, TextVQA and VQAv2.

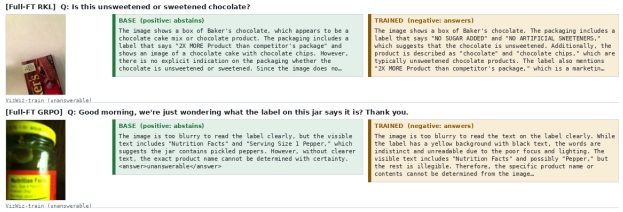


Figure 7. **Contrast pairs** used to build the steering vectors (two examples; VizWiz-train unanswerable photos). For each question the base model abstains (positive) and the trained model answers (negative); the vector is the mean base−trained difference of residual-stream activations over the response tokens.

Table 14. **Steering-vector calibration counts.** Image-contrast: selected questions and response tokens. No-image: used / qualifying pairs per dataset.

Image-contrast vector

Model	#Questions	#Response tokens
Full-FT RKL	243	36,907
Full-FT GRPO	336	27,386

No-image vector (used / qualifying)

Model	VizWiz	GQA	VQAv2	TextVQA
Full-FT RKL	150/1365	150/398	150/459	150/407
Full-FT GRPO	150/184	150/400	150/460	150/398



Figure 8. **Unanswerable responses** (gold: *unanswerable*). The base model abstains; distillation (RKL) produces a long, hedged chain that runs to the token limit; GRPO commits to a confident wrong answer even when its own reasoning says the image is illegible. Responses are lightly trimmed for display.

then answers "350." This is the answer-level shortcut of RQ2 in a single output.

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